

The flushing toilet

By washing away disease-causing sewage, the flushing toilet has saved millions of lives. Surprisingly, it didn't immediately catch on when it was invented in the 16th century. The flushing mechanism wouldn't achieve mass popularity until someone came up with a way of removing not just the waste, but also the smells.



JOHN HARRINGTON

The English poet John Harrington invented the first flushing toilet in 1596. It had a cistern that released water into a toilet bowl, washing its contents into a pit below. Despite making a version for his godmother, Queen Elizabeth I, Harrington's toilet didn't prove popular.

THE S-BEND TOILET

The problem with early flushing toilets was that bad smells could travel up through the pipes that took away the waste. In 1775, the Scottish inventor Alexander Cummings came up with a solution in the form of an S-shaped water trap, which prevented bad smells from escaping.



S-bend water trap

S-bend toilet, 1870



Sir Joseph Bazalgette (below center) inspects the progress of London's new sewage system, 1860s

SEWAGE SYSTEMS

The widespread use of flushing toilets in the mid-19th century prompted the construction of new sewage systems to wash away the waste. An extensive system was built below London, UK, by the British engineer Sir Joseph Bazalgette. It made the city a cleaner and safer place, helping to end the spread of deadly diseases such as cholera.

The flush is activated by pulling a chain.

